OpenClaw Camera Application Tutorial

> Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. This article assumes the basic environment is already prepared and only covers camera-related content.

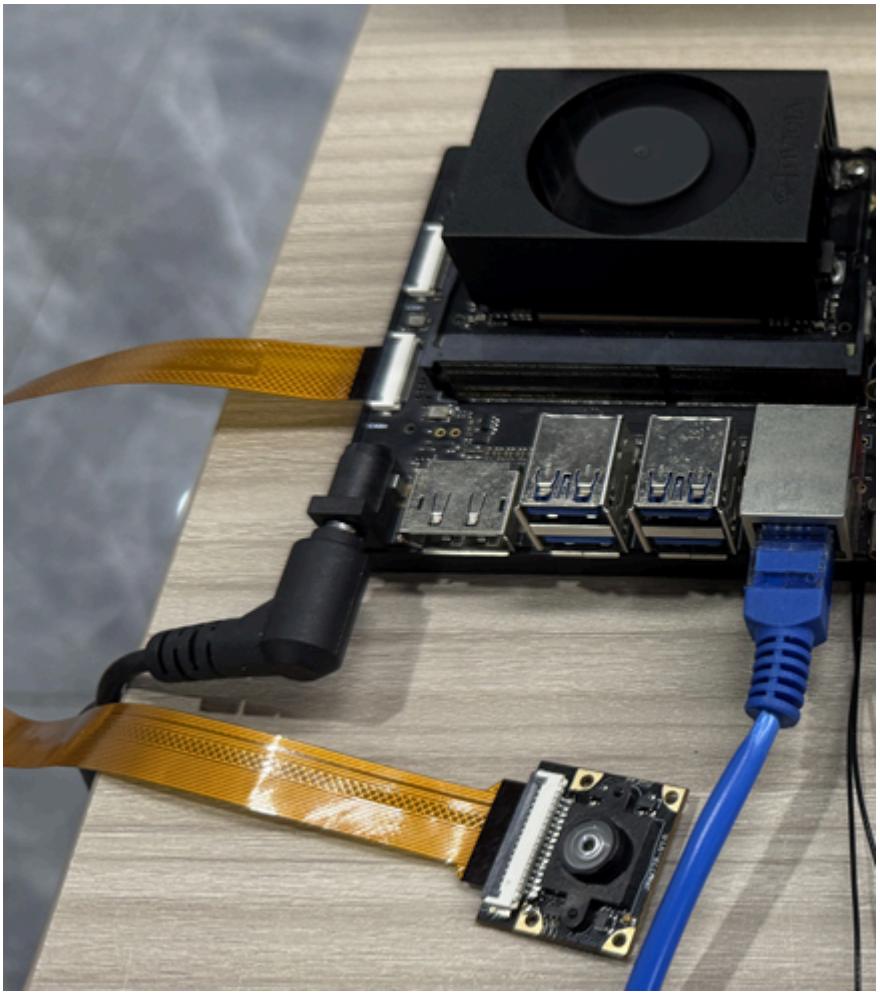
1. Tutorial Objective

This article demonstrates OpenClaw's natural language control capabilities through the CSI camera peripheral. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw using the camera for photo capture and visual analysis
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four camera cases: photo archiving, scene description, OCR recognition, object positioning and face detection
- Understand the complete chain from "user submitting a visual task" to "image capture" and "result return"

2. Hardware Preparation

- OpenClaw main control board
- CSI camera module
- Camera ribbon cable
- `[Optional] Microphone module`
- Wiring diagram



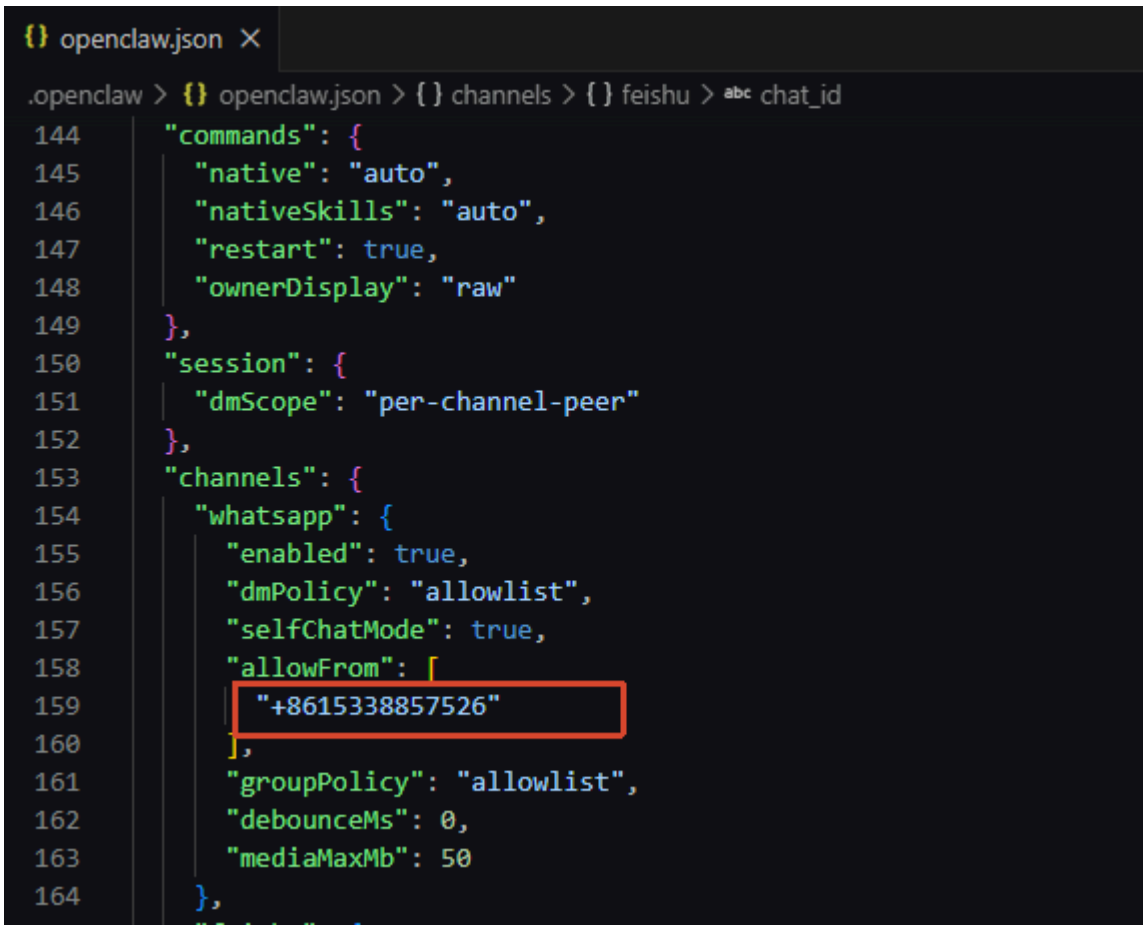
3. Software Preparation

If using WhatsApp, to upload images directly to WhatsApp, you need to enter your WhatsApp dialogue ID in the `allowFrom` field at the following path

- Open your newly created WhatsApp dialogue to see the dialogue ID



- Fill this into "allowFrom":



- Enter the following command in the terminal to apply

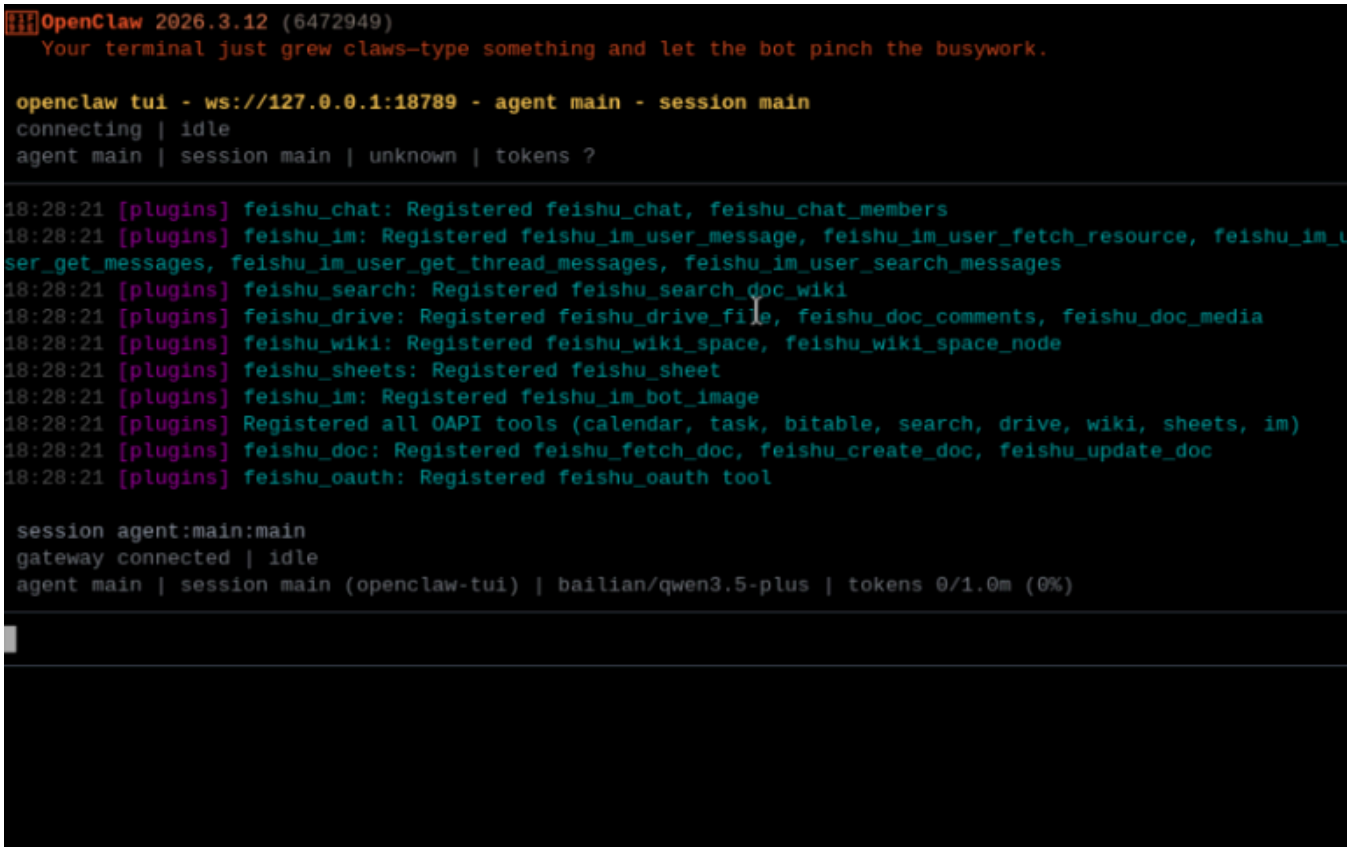
```
openclaw gateway restart
```

4. Testing Camera Application

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests]. Also, please confirm that the visual model has been configured, refer to [02-OpenClaw API-KEY Configuration] and [03-OpenClaw Model Switching]. After passing, proceed to the camera-specific test.

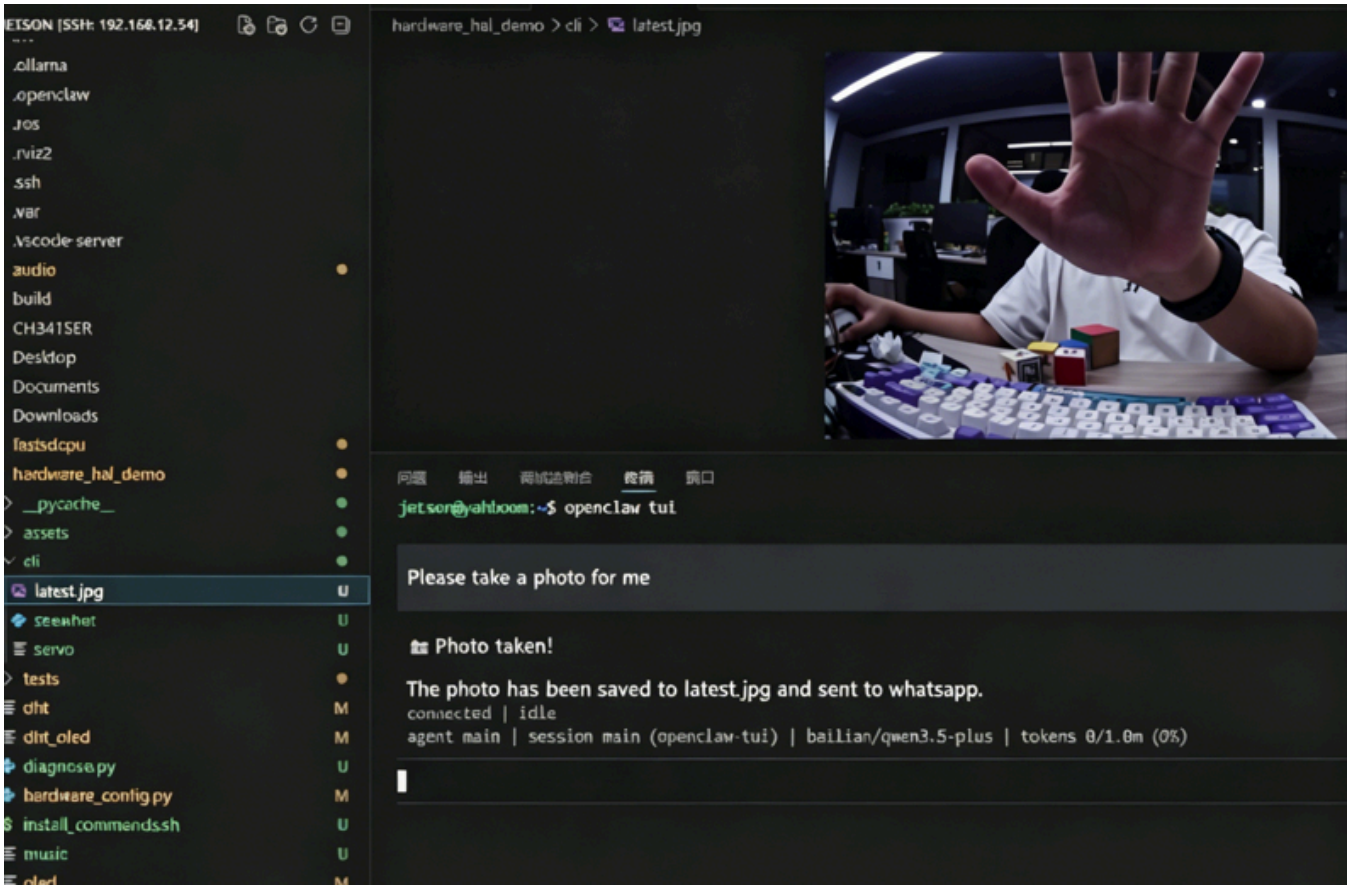
Enter the following command in the terminal to enter TUI:

```
openclaw tui
```



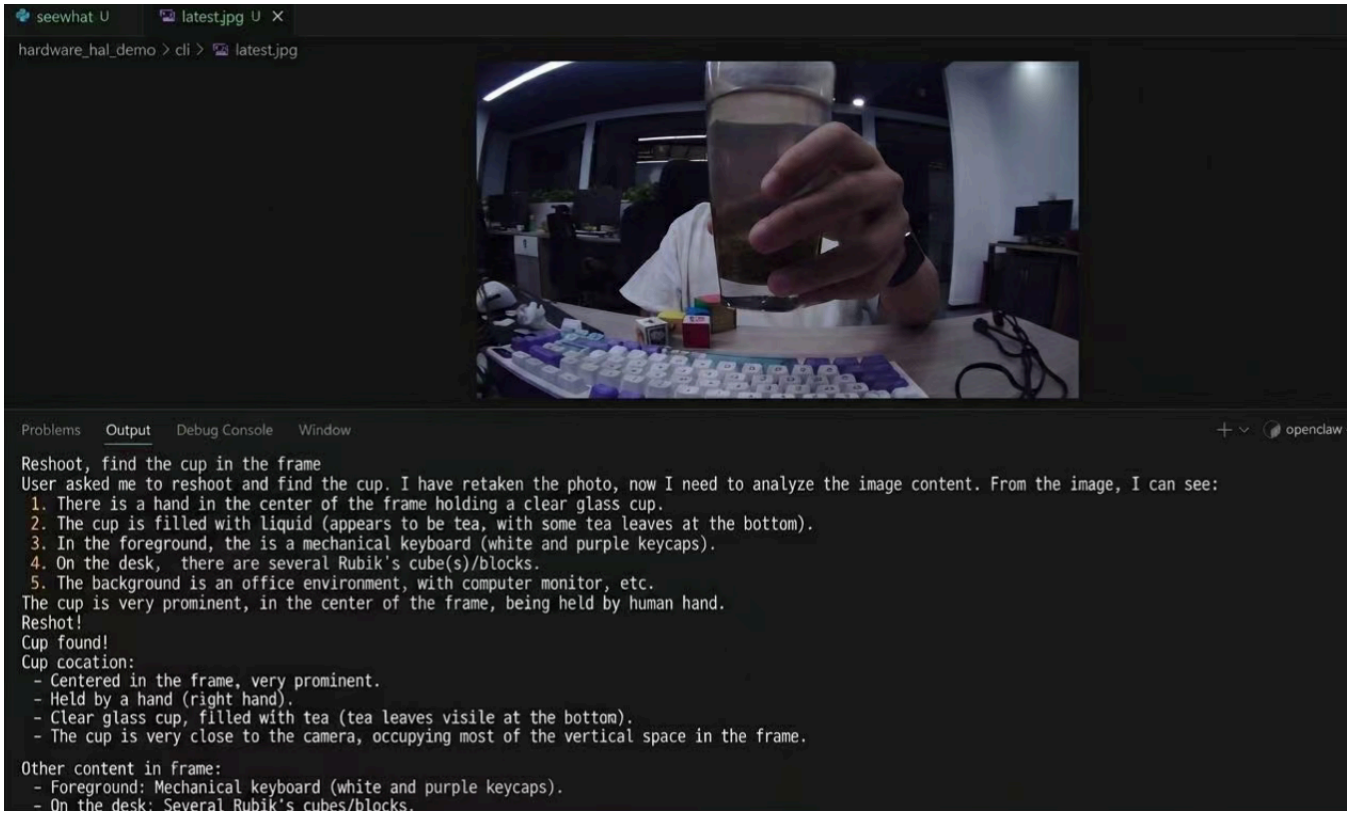
You can try entering the following content in the terminal:

- Help me take a photo
- Recognize the text in this image

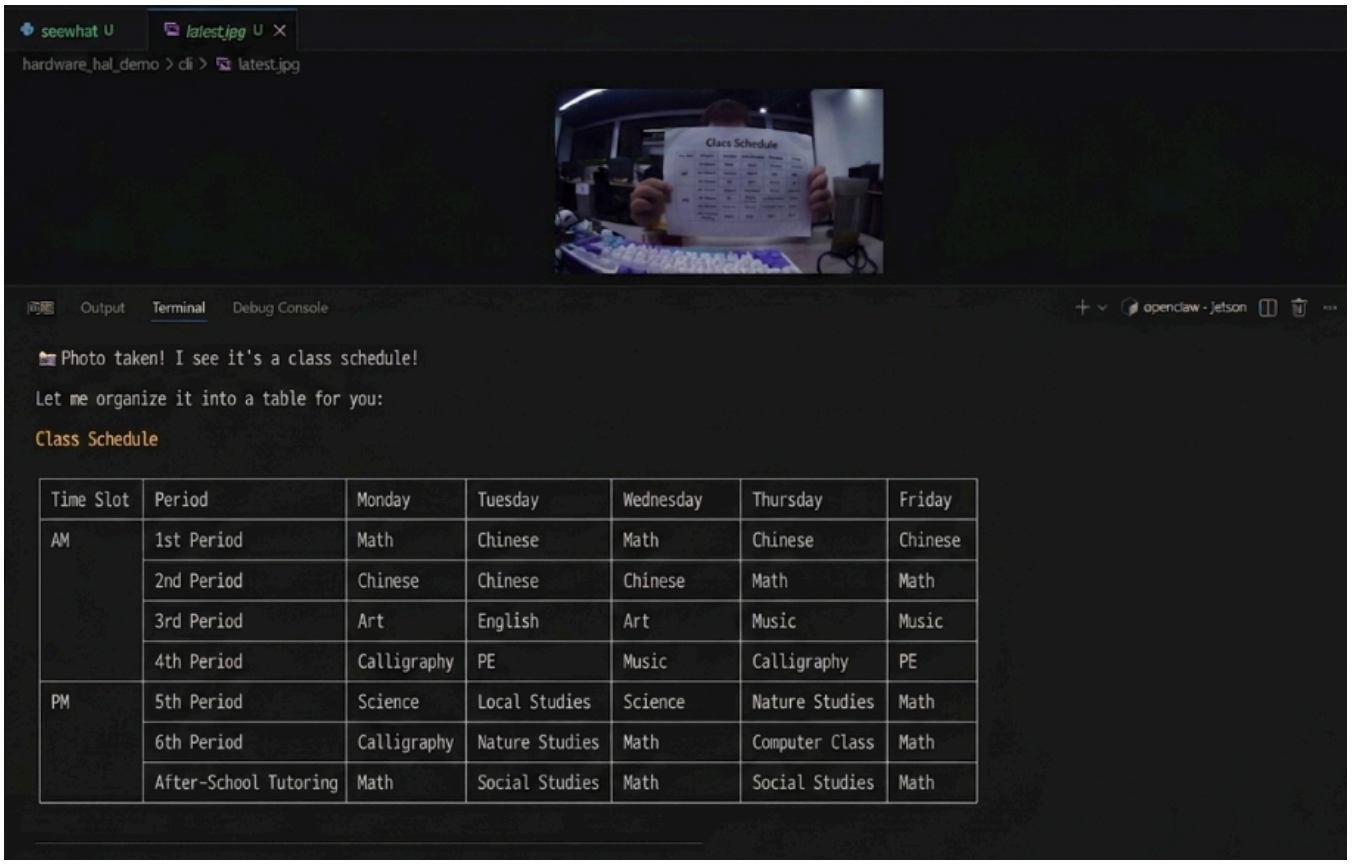


- Find the cup in the frame (because the model will reference context, you need to remind it to retake the

photo, otherwise it might use the first photo)



- Retake a photo and organize the text in it into a table



5. Three Schemes for Communicating with OpenClaw

Scheme	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote control, classroom demonstration	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the visual chain
Voice code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration scheme

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.

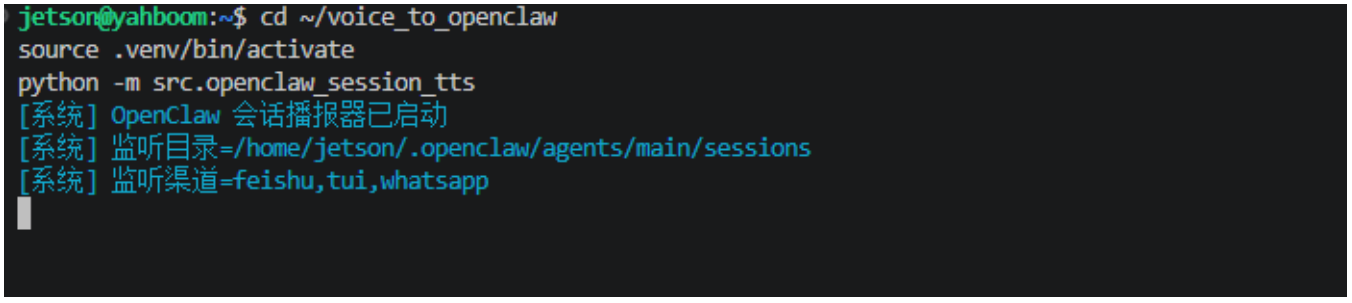
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



5.1 whatsapp Scheme

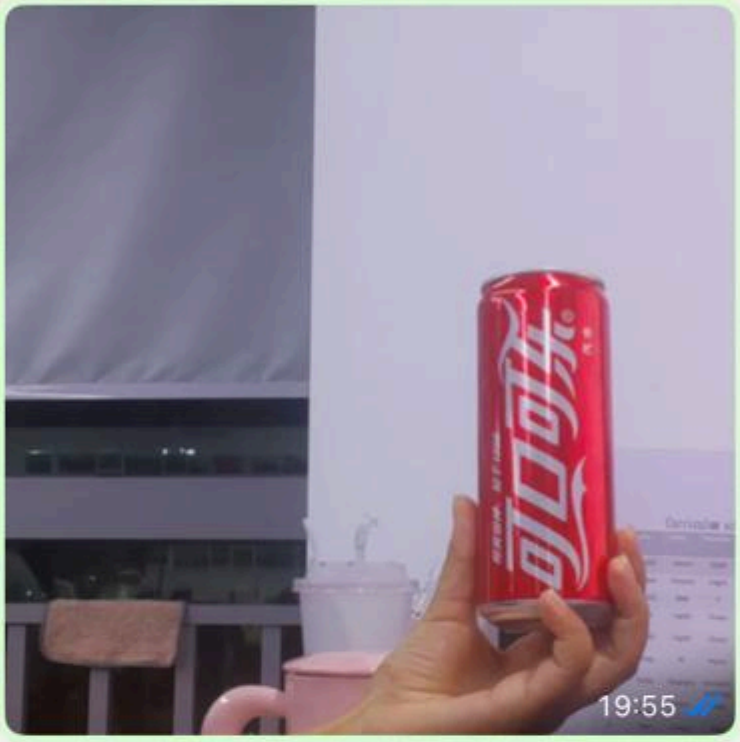
For WhatsApp connection steps, please directly refer to [06-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the camera tutorial.

After opening WhatsApp, you can usually also see the photos and recognition results triggered by `openclaw tui` earlier synchronized over.

19:55

< 2 +86 153 3885 7526 (你)
给自己发消息

Look at what's in front of 19:55



I can see the scene in front of the camera:

- A hand is holding a **red Coca-Cola can** in the center
- To the right: a small **green desk fan** and a **water bottle**
- In the background: a **curriculum schedule** paper posted on the wall
- On the left: what looks like a **purple phone case** and a **window with blinds**
- Various items on the desk including a **pink mug** and some containers

The scene appears to be a desk or



5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue the conversation:

```
openclaw tui
```

Then you can directly enter camera-related commands, for example:

- Take a photo of the current scene
- Describe what the camera sees
- Check if there are any faces

You can add your own TUI dialogue screenshots here.

5.3 Voice Code Scheme

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [05-AI Voice Interaction Configuration]. After wake-up, you can issue voice commands directly, for example:

- Help me take a photo
- Look at what's ahead
- Help me read the text in the frame

You can add your own voice wake-up and voice control screenshots here.

6. Comprehensive Cases

Note: The following are all demonstrated using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Photo Archiving

Experimental Objective

Have OpenClaw first complete the most basic photo-taking function and save the image.

Effect Description

After execution, a photo will be generated locally, typically saving both the original photo file and a `latest.jpg` alias for the most recent photo. This case is suitable for verifying the image capture chain.

Example Instructions

- Help me take a photo
- Capture the current scene
- Take a photo now for records

6.2 Case 2: Scene Description

Experimental Objective

Have OpenClaw provide a brief description of the current scene after taking a photo.

Effect Description

OpenClaw can first complete the photo capture, then call the visual analysis script to provide a one-sentence description of the image content, such as identifying desktops, people, food, or common objects.

Example Instructions

- Look at what's ahead
- Help me describe the current scene
- After taking a photo, tell me what's mainly in the frame

6.3 Case 3: OCR Table Recognition

Experimental Objective

Have OpenClaw recognize text in images and organize it into a table.

Effect Description

This case is suitable for photographing labels, cards, packaging, or simple documents, then extracting the text. It can output plain text or continue to organize it into a table.

Example Instructions

- Recognize the text in this image
- After taking a photo, organize the text into a table
- Find the cup in the frame
- Check if there are any faces
- Help me locate the phone on the right side of the desk